2013 Homework 3

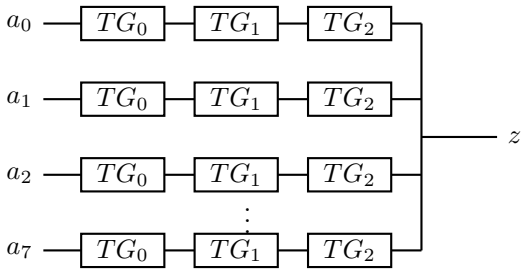
Fixed-size transmission-gate MUX and inverter short-circuit power

The original printed prompt is preserved at `../assignments/homework3.pdf`. Both supplied DOCX versions of the 100501015 answer were read, including their embedded plots. The original 10 fF load and fixed TG dimensions are followed here.

1. Baseline 8:1 transmission-gate multiplexer

Set $S = s_0 + 2s_1 + 4s_2$ and require $z = a_S$. Every TG uses the specified $W_n = 10\,\mu$m, $W_p = 20\,\mu$m and $L = 0.18\,\mu$m. It contains parallel NMOS/PMOS channels, enabled by complementary gates. Generate each $\bar{s}_i$ once with a shared CMOS inverter. All PMOS bodies connect to VDD and NMOS bodies to ground; the load is exactly 10 fF.

The baseline provides eight separate three-TG paths, each decoding one select word. Its 24 TGs plus three select inverters require $48 + 6 = 54$ MOSFETs. The selected path crosses three TGs; inactive paths add capacitance at their shared output. Use the same 10 fF output load and 20 ps data slew for both designs.

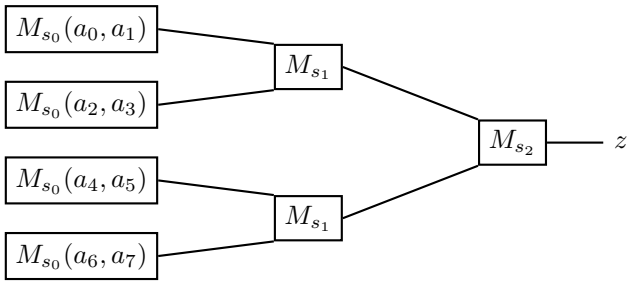


Path j: each TG selects bit j_k of s_k

The diagram abbreviates the middle four paths. Their complete connections, dimensions and control polarities are in the standalone `mux.sp`.

Improved topology with unchanged TG sizes

Replace the separate paths by a binary 2:1 tree: four first-stage muxes select adjacent data pairs with s_0; two second-stage muxes use s_1; one final mux uses s_2. Each 2:1 mux uses two TGs. Thus the improved network has 14 TGs and three shared inverters, or $28 + 6 = 34$ MOSFETs. The depth remains three TGs, but only two final TG terminals load the output instead of eight.



Each M_s passes its lower-index data when $s = 0$ and its higher-index data when $s = 1$. No TG transistor is resized to obtain the improvement.

Actual HSPICE delay comparison

Conditions: CIC018 TT, 1.8 V, 25°C, 10 fF output load, 20 ps input 0–100% slew, 1 ps maximum timestep. Each of eight settled select words is tested with the selected data rising and falling, all other data held low. Both designs run in the same sweep with independent loads. Propagation delay is the last output 0.9 V crossing minus the input 0.9 V crossing.

Select	Baseline LH ps	Baseline HL ps	Tree LH ps	Tree HL ps
0	212.257	203.307	66.352	76.191
1	213.822	201.262	68.550	73.761
2	215.152	201.274	68.285	74.978
3	216.681	199.180	70.416	72.510
4	216.298	200.784	67.063	75.130
5	217.825	198.714	69.242	72.682
6	219.019	198.684	68.982	73.898
7	220.507	196.567	71.084	71.416

$$\boxed{t_{\max,baseline} = 220.507 \text{ ps}, \quad t_{\max,tree} = 76.191 \text{ ps}.}$$

The measured worst-case reduction is 65.447%. Every one of the 16 paired transition tests shows an improvement. Before/after levels pass low ≤ 0.2 V and high ≥ 1.6 V. These maxima apply to the declared data-edge tests; select-switching hazards and PVT extremes are not claimed to be covered. An independent Boolean tree evaluation checks all $8 \times 256 = 2048$ legal select/data combinations. Its truth evidence is not mislabeled as 2048 physical HSPICE experiments.

Reference comparison. The supplied answer correctly proposes a shared binary tree. However, its baseline lists a 1 fF load rather than the printed 10 fF; the improved code omits the load, and its data slew changes from 200 ps to 100 ps. Those experiments cannot establish a matched-condition speedup. We retain the tree idea, share select complements, use the same required load and stimuli, and measure both directions for every select word. Reported reference delays 241.765/78.465 ps are not reused as our measurements.

Reproduce and evidence

Run `python reproduce.py` with `-model D:/cic018.l` and `-work PATH`. The optional `-hspice` flag selects another installation. `-prepare-only` regenerates the 2048-case logic table and decks without claiming a simulator run. The reusable circuit is `mux.sp`; both testbenches use the local process library. `results/summary.json`, `mux_delays.csv`, `inverter_power.csv` and `logic_2048.csv` preserve the checks. Raw simulator and license logs remain outside the repository. No process model or layout artifacts are published.

2. Short-circuit power versus load capacitance

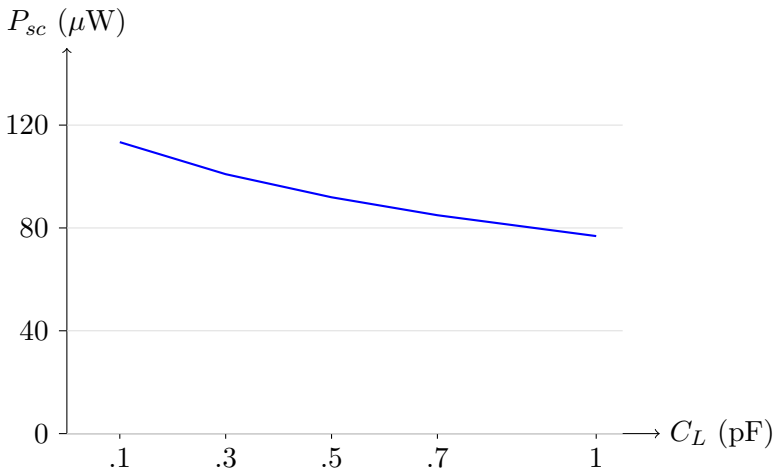
Choose inverter $W_p = 20\,\mu$m, $W_n = 10\,\mu$m, $L = 0.18\,\mu$m; its sizes are not specified by this part of the prompt. Use 1.8 V, TT, 25°C, the required 3 ns rising and falling input ramps, and a declared 20 ns period. The corrected single capacitor is parameterized at the inverter's actual output for $C_L = 0.1, 0.3, 0.5, 0.7, 1.0$ pF.

During an input transition, NMOS and PMOS can both conduct. Separately from capacitive charging, estimate their common through-channel current using HSPICE's intrinsic channel-current outputs, LX4:

$$I_{sc}(t) = \max\{0, \min[I_{p,ch}(t), I_{n,ch}(t)]\}, \qquad P_{sc} = \frac{V_{DD}}{T}\int_{t_0}^{t_0+T} I_{sc}(t)\,dt.$$

LX4 is positive for the normally biased PMOS and NMOS here; the outer clamp removes reverse-bias/overshoot values. This is an explicitly declared channel-overlap estimator, not total transistor dissipation, terminal displacement current, an edge-window average, or peak power. Average a complete settled cycle from 23 to 43 ns. Supply power is measured independently as $\langle -V_{DD} I(VDD)\rangle$. External-load switching power is $C_L V_{DD}^2/T$.

C_L (pF)	P_{sc} (μW)	Supply power (μW)	$C_L V_{DD}^2/T$ (μW)
0.1	113.373082	138.184419	16.200000
0.3	100.912372	158.139075	48.600000
0.5	91.925520	181.573889	81.000000
0.7	84.947117	207.021195	113.400000
1.0	76.828513	247.545707	162.000000



The measured overlap power decreases as the load grows for this fixed input slew and sizing. During the input's overlap interval, a larger load keeps the output closer to its previous rail, reducing the drain-source voltage of one of the conducting devices and limiting direct through current. Meanwhile, external-load switching power increases linearly with C_L; total supply power must therefore be distinguished from P_{sc}. A universal monotonic law for arbitrary slew, size and activity is not claimed.

Reference correction. The supplied inverter uses node `out`, but its altered capacitors are connected to `vout`; the original 0.1 pF capacitor remains unchanged. It also averages `P(MOS)` over changing edge windows. Thus its load sweep and listed powers do not reproduce the required cycle-average short-circuit curve. The corrected experiment changes one actual load and records intrinsic-current overlap and supply power separately. All five sampled high/low output levels are valid.